Supervised Bayesian Matrix Factorization

YFB Parameterization, $\boldsymbol{\eta} = (YF)\boldsymbol{\beta}$: Full Derivation and Update Algorithm

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1 Introduction

This document derives the coordinate-ascent variational inference (CAVI) algorithm for the YFB parameterization of supervised Bayesian matrix factorization. It follows the same path as the LB document (`SSBMF-LB-model-derivations/MF_Derivations_UpdateAlgo_REVISED.pdf`): the joint model, the objective, the Taylor approximation of the Cox partial likelihood, one update per parameter, and a one-page algorithm summary (Section 12).

The two parameterizations differ in one place. In LB the Cox linear predictor is $\eta_i = \boldsymbol{l}_i^\top \boldsymbol{\beta}$, so the patient scores L carry the survival information. In YFB the linear predictor is $\eta_i = \boldsymbol{y}_i^\top F \boldsymbol{\beta}$, the patient's expression profile projected onto the gene loadings. The gene loadings F are now the parameter that appears in both likelihoods, and L enters only the reconstruction.

Every update reduces to an empirical Bayes normal means (EBNM) problem: a pseudo-observation $x = B/A$ with standard error $s = 1/\sqrt{A}$, where A collects the coefficients of the second moment and B the coefficients of the first moment in the expected log-likelihood. Solving it with the `ebnm` package returns the posterior moments and the fitted prior.

Configuration documented. The derivation describes `fit_ssbmf_yfb(canonicalize = TRUE, w_S =` w_S`)`, the configuration used for the supervision studies from October 2026 onward (point-Laplace priors on L and F, a normal prior on each β_k). The earlier benchmark configuration (`alpha_F = 0`) is the special case of Section 15. Options that were tested but not adopted are in Appendix A. Differences from the April 2026 derivations are listed in Section 17.

1.1 Update Procedure

For $b = 1, 2, \ldots$ until convergence:

1. Compute the Cox Taylor working quantities W_{ii} and z_i at the current linear predictor.
2. For $k = 1, \ldots, K$:
   (a) Update $q(\beta_k)$, $g(\beta_k)$ (survival term only).
   (b) Update $q(l_k)$, $g(l_k)$ (genomics term only: the standard EBMF update).
   (c) Update $q(f_k)$, $g(f_k)$ (genomics and survival terms: differs from EBMF).
3. Update τ_j (the standard EBMF update).
4. Rescale each program so its training scores have unit standard deviation (Section 9).

2 Notation and Model Setup

2.1 Index Conventions

Symbol	Dimension	Meaning
$n,\ p,\ K$	scalars	patients, genes, programs (latent factors)
$i,\ j,\ k$		patient, gene and program indices
Y	$n \times p$	expression, column-centered and standardized within platform
L	$n \times K$	patient scores; l_{ik} = score of patient i on program k
F	$p \times K$	gene loadings; f_{jk} = loading of gene j on program k
$Z = YF$	$n \times K$	program scores; $z_{ik} = \sum_j y_{ij} f_{jk}$
$\boldsymbol{\beta}$	$K \times 1$	Cox coefficients
τ	$p \times 1$	gene-specific noise precision
(t_i, δ_i)		survival time and event indicator
D	scalar	number of events, $\sum_i \delta_i$
w_S	scalar	survival weight (Section 3)

2.2 Posterior Moment Notation

Bars denote moments under the variational posterior q:

$$\bar{l}_{ik} = \mathbb{E}_q[l_{ik}], \qquad \overline{l^2_{ik}} = \mathbb{E}_q[l^2_{ik}] = \bar{l}^2_{ik} + \mathrm{Var}_q(l_{ik}),$$

and likewise $\bar{f}_{jk}$, $\overline{f^2_{jk}}$, $\bar{\beta}_k$, $\overline{\beta^2_k}$.

2.3 The Program Scores $Z = YF$

Z is not a separate parameter. It is a fixed linear function of the observed expression Y and the loadings F, so its distribution under q is induced by $q(F)$. Under the mean-field posterior the loadings are independent across genes and programs, which gives

$$\bar{z}_{ik} = \sum_j y_{ij}\, \bar{f}_{jk}, \qquad V^Z_{ik} := \mathrm{Var}_q(z_{ik}) = \sum_j y^2_{ij}\, \mathrm{Var}_q(f_{jk}), \qquad \overline{z^2_{ik}} = \bar{z}^2_{ik} + V^Z_{ik}, \tag{1}$$

and z_{ik} and $z_{ik'}$ are independent for $k \neq k'$.

No prior is placed on Z

The prior is on F. Because Z is a deterministic function of F given Y, its posterior moments follow from $q(F)$ through Eq. (1). The question of which prior is induced on Z arises only in a model where Z replaces L as a free parameter, and this model does not do that.

2.4 Models

Genomics model.

$$Y = LF^\top + E, \qquad E_{ij} \overset{\text{ind}}{\sim} \mathcal{N}(0, \tau_j^{-1}). \tag{2}$$

Survival model.

$$h_i(t) = h_0(t)\exp(\eta_i), \qquad \eta_i = \boldsymbol{y}_i^\top F\boldsymbol{\beta} = \sum_k z_{ik}\beta_k, \tag{3}$$

with h_0 unspecified and the Cox partial likelihood (Breslow handling of ties). β_k is the log hazard ratio per unit of program k's score; after the scaling of Section 9 that unit is one training standard deviation.

Priors.

$$l_{ik} \overset{\text{iid}}{\sim} g_{l_k}, \qquad f_{jk} \overset{\text{iid}}{\sim} g_{f_k}, \qquad \beta_k \sim g_{\beta_k},$$

each estimated by empirical Bayes. In the documented configuration g_{l_k} and g_{f_k} are point-Laplace (point-exponential gives nonnegative programs) and g_{β_k} is $\mathcal{N}(0, \sigma_k^2)$ with σ_k^2 estimated.

Joint likelihood.

$$p(Y, t, \delta \mid L, F, \boldsymbol{\beta}, \tau) = \underbrace{p(Y \mid L, F, \tau)}_{\text{genomics}}\ \underbrace{p(t, \delta \mid Y, F, \boldsymbol{\beta})}_{\text{survival}}. \tag{4}$$

The survival model conditions on the observed Y, which is a covariate in the Cox model as well as the outcome of the factorization. In LB the survival model conditions on the latent L instead.

3 Objective Function

The objective, maximized over $q = \prod_k q_{l_k} q_{f_k} q_{\beta_k}$, the priors and τ, is

$$\begin{aligned}\mathcal{F}(q, g, \tau) = {} & \mathbb{E}_q[\log p(Y \mid L, F, \tau)] + w_S\, \mathbb{E}_q[\ell_{\text{Cox}}(YF\boldsymbol{\beta})] \\ & - \sum_k \mathrm{KL}(q_{l_k} \| g_{l_k}) - \sum_k \mathrm{KL}(q_{f_k} \| g_{f_k}) - \sum_k \mathrm{KL}(q_{\beta_k} \| g_{\beta_k}).\end{aligned} \tag{5}$$

The survival weight w_S

w_S multiplies the Cox log-likelihood. At $w_S = 1$, Eq. (5) is the ordinary evidence lower bound and its maximizer approximates the Bayesian posterior. At $w_S \neq 1$ it is the lower bound for a power (generalized-Bayes) posterior in which the survival likelihood is raised to the power w_S. The genomics term keeps weight 1. The same w_S appears in every update that involves survival and in the monitored objective. At $w_S = 0$ survival drops out of every update, including β (Section 15).

Genomics term. With $\bar{R}_{ij}^2 := \mathbb{E}_q[(y_{ij} - \boldsymbol{l}_i^\top \boldsymbol{f}_j)^2]$,

$$\mathbb{E}_q[\log p(Y \mid L, F, \tau)] = \sum_j \Big[\tfrac{n}{2}\log\tau_j - \tfrac{\tau_j}{2}\sum_i \bar{R}_{ij}^2\Big] + C, \tag{6}$$

$$\bar{R}_{ij}^2 = \Big(y_{ij} - \sum_k \bar{l}_{ik}\bar{f}_{jk}\Big)^2 + \sum_k (\overline{l_{ik}^2 f_{jk}^2} - \bar{l}_{ik}^2 \bar{f}_{jk}^2).$$

4 Taylor Approximation of the Cox Partial Likelihood

The Cox partial log-likelihood $\ell(\boldsymbol{\eta})$ is not conjugate to any of the priors. As in the LB document (Section 3, which shows every step), it is replaced by its second-order expansion around the current linear predictor $\hat{\boldsymbol{\eta}} = Y\bar{F}\bar{\boldsymbol{\beta}}$, keeping the diagonal of the Hessian:

$$\ell(\boldsymbol{\eta}) \approx \ell(\hat{\boldsymbol{\eta}}) - \frac{1}{2}\sum_i W_{ii}\,(z_i - \eta_i)^2 + C, \qquad z_i = \hat{\eta}_i + \frac{u_i}{W_{ii}}. \tag{7}$$

Here z_i is the working response. With $\hat{\theta}_i = e^{\hat{\eta}_i}$, risk-set sums $\hat{S}(t) = \sum_{l:t_l \geq t} \hat{\theta}_l$ and Breslow cumulative hazard $\hat{H}(t) = \sum_{\text{event times } s \leq t} d(s)/\hat{S}(s)$, the score and the weight are

$$u_i = \delta_i - \hat{\theta}_i \hat{H}(t_i), \qquad W_{ii} = \hat{\theta}_i \hat{H}(t_i). \tag{8}$$

Implementation Note

The exact diagonal of the negative Hessian is $\hat{\theta}_i \hat{H}(t_i) - \hat{\theta}_i^2 \sum_{s \leq t_i} d(s)/\hat{S}(s)^2$. The code (`calc_cox_taylor_yf()`) drops the second term, which is small relative to the first when risk sets are large, and floors W_{ii} at 10^{-6}. Because the scores sum to zero at any $\hat{\boldsymbol{\eta}}$, $\sum_i W_{ii} = \sum_i \hat{\theta}_i \hat{H}(t_i) = D$ before the floor: the total survival weight equals the number of events. The expansion is computed once per outer iteration (Step 1 of the algorithm), as in LB.

Expected surrogate under q. Taking expectations of Eq. (7),

$$\mathbb{E}_q[\ell(\boldsymbol{\eta})] \approx \ell(\hat{\boldsymbol{\eta}}) - \frac{1}{2}\sum_i W_{ii}\Big[(z_i - \bar{\eta}_i)^2 + \mathrm{Var}_q(\eta_i)\Big] + C, \qquad \mathrm{Var}_q(\eta_i) = \sum_k \big(\overline{z_{ik}^2 \beta_k^2} - \bar{z}_{ik}^2 \bar{\beta}_k^2\big), \tag{9}$$

since $z_{ik}\beta_k$ are independent across k under the mean-field posterior.

Partial working response. For program k, remove the other programs' contributions:

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{z}_{ik'}\,\bar{\beta}_{k'}. \tag{10}$$

5 Update for $q(\beta_k)$: Survival Coefficients

Only the survival term and $\mathrm{KL}(q_{\beta_k} \| g_{\beta_k})$ involve β_k.

Derivation Steps

Write $\eta_i = z_{ik}\beta_k + \sum_{k' \neq k} z_{ik'}\beta_{k'}$. Keeping the terms of $-\frac{1}{2}\mathbb{E}_q[(z_i - \eta_i)^2]$ that involve β_k, and using the independence of β_k, z_{ik} and the other programs under q:

$$\begin{aligned} -\tfrac{1}{2}\mathbb{E}_q\big[(z_i - \eta_i)^2\big] &= -\tfrac{1}{2}\Big[\mathbb{E}_q[\beta_k^2]\,\mathbb{E}_q[z_{ik}^2] - 2\,\mathbb{E}_q[\beta_k]\,\bar{z}_{ik}\Big(z_i - \sum_{k' \neq k} \bar{z}_{ik'}\bar{\beta}_{k'}\Big)\Big] + C \\ &= -\tfrac{1}{2}\,\overline{\beta_k^2}\,\overline{z_{ik}^2} + \bar{\beta}_k\,\bar{z}_{ik}\,z_i^{-k} + C. \end{aligned}$$

Multiply by $w_S W_{ii}$ and sum over i. The result has the form $-\frac{1}{2}A\,\mathbb{E}_q[\beta_k^2] + B\,\mathbb{E}_q[\beta_k]$, the expected log-likelihood of a single normal observation of β_k.

$$A_{\beta_k} = w_S \sum_i W_{ii}\,\overline{z_{ik}^2}, \qquad B_{\beta_k} = w_S \sum_i W_{ii}\, z_i^{-k}\,\bar{z}_{ik}. \tag{11}$$

EBNM Problem

$$(\bar{\beta}_k,\, \overline{\beta_k^2},\, \hat{g}_{\beta_k}) \;\leftarrow\; \text{EBNM}\left(x_k = \frac{B_{\beta_k}}{A_{\beta_k}},\; s_k = \frac{1}{\sqrt{A_{\beta_k}}}\right).$$

w_S cancels in x_k and scales s_k by $w_S^{-1/2}$: a larger weight narrows the survival likelihood for β_k relative to its prior. The term V_{ik}^Z inside $\overline{z_{ik}^2}$ is the error-in-variables correction for uncertainty in the loadings. The prior $g_{\beta_k} = \mathcal{N}(0, \sigma_k^2)$ is refit at every update from this single pseudo-observation, which gives $\hat{\sigma}_k^2 = \max(0, x_k^2 - s_k^2)$.

6 Update for $q(l_k)$: Patient Scores

L does not appear in the Cox model, so only the genomics term and $\mathrm{KL}(q_{l_k} \| g_{l_k})$ involve l_k. With the partial residual

$$R_{ij}^{-k} = y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}, \tag{12}$$

the terms of Eq. (6) that involve l_{ik} are

$$-\tfrac{1}{2} \sum_j \tau_j \left[\mathbb{E}_q[l_{ik}^2]\,\overline{f_{jk}^2} - 2\,\mathbb{E}_q[l_{ik}]\, R_{ij}^{-k}\, \bar{f}_{jk}\right] + C,$$

which gives

$$A_{l_k} = \sum_j \tau_j\, \overline{f_{jk}^2} \quad \text{(the same for every patient)}, \qquad B_{l_{ik}} = \sum_j \tau_j\, R_{ij}^{-k}\, \bar{f}_{jk}. \tag{13}$$

EBNM Problem

$$(\bar{l}_{\cdot k},\, \overline{l_{\cdot k}^2},\, \hat{g}_{l_k}) \;\leftarrow\; \text{EBNM}\left(x_i = \frac{B_{l_{ik}}}{A_{l_k}},\; s = \frac{1}{\sqrt{A_{l_k}}}\right), \qquad i = 1, \ldots, n.$$

This is the standard EBMF update for scores. Survival reaches L only through F: the current $\bar{f}_{jk}$ and $\overline{f_{jk}^2}$ carry whatever survival information the F update put into them. g_{l_k} is refit at every update.

7 Update for $q(f_k)$: Gene Loadings

f_{jk} appears in both likelihoods: in the reconstruction through $\boldsymbol{l}_i^\top \boldsymbol{f}_j$, and in the Cox model through $z_{ik} = \sum_j y_{ij} f_{jk}$. F is the dual-source parameter of YFB, as L is in LB.

7.1 Genomics Contribution

The terms of Eq. (6) that involve f_{jk} are $-\frac{\tau_j}{2}\sum_i\big[\overline{l_{ik}^2}\,\mathbb{E}_q[f_{jk}^2] - 2\,R_{ij}^{-k}\bar{l}_{ik}\,\mathbb{E}_q[f_{jk}]\big]$, so

$$A_{f_{jk}}^{\text{gen}} = \tau_j \sum_i \overline{l_{ik}^2}, \qquad B_{f_{jk}}^{\text{gen}} = \tau_j \sum_i R_{ij}^{-k}\,\bar{l}_{ik}. \tag{14}$$

7.2 Survival Contribution

Derivation Steps

Separate gene j from the other genes of program k:

$$\eta_i = \beta_k\, y_{ij} f_{jk} + \beta_k\, s_{ij} + \eta_i^{-k}, \qquad s_{ij} = \sum_{j'\neq j} y_{ij'} f_{j'k}, \qquad \eta_i^{-k} = \sum_{k'\neq k} z_{ik'}\beta_{k'}.$$

Expand $-\frac{1}{2}\mathbb{E}_q[(z_i-\eta_i)^2]$ and keep the terms that involve f_{jk}. Under the mean-field posterior f_{jk}, β_k, s_{ij} and η_i^{-k} are independent, so

$$\begin{aligned}
-\tfrac{1}{2}\mathbb{E}_q\big[(z_i-\eta_i)^2\big] &= -\tfrac{1}{2}\Big[\overline{\beta_k^2}\,y_{ij}^2\,\mathbb{E}_q[f_{jk}^2] + 2\,\overline{\beta_k^2}\,y_{ij}\,\bar{s}_{ij}\,\mathbb{E}_q[f_{jk}] \\
&\qquad + 2\,\bar{\beta}_k\,y_{ij}\,\bar{\eta}_i^{-k}\,\mathbb{E}_q[f_{jk}] - 2\,z_i\,\bar{\beta}_k\,y_{ij}\,\mathbb{E}_q[f_{jk}]\Big] + C \\
&= -\tfrac{1}{2}\,\overline{\beta_k^2}\,y_{ij}^2\,\mathbb{E}_q[f_{jk}^2] + \Big[\bar{\beta}_k\,y_{ij}\,z_i^{-k} - \overline{\beta_k^2}\,y_{ij}\,\bar{s}_{ij}\Big]\mathbb{E}_q[f_{jk}] + C,
\end{aligned}$$

using $z_i - \bar{\eta}_i^{-k} = z_i^{-k}$ (Eq. 10) and $\bar{s}_{ij} = \bar{z}_{ik} - y_{ij}\bar{f}_{jk}$. Multiply by $w_S W_{ii}$ and sum over i.

$$A_{f_{jk}}^{\text{surv}} = \overline{\beta_k^2}\sum_i W_{ii}\,y_{ij}^2, \qquad B_{f_{jk}}^{\text{surv}} = \bar{\beta}_k\sum_i W_{ii}\,y_{ij}\,z_i^{-k} - \overline{\beta_k^2}\sum_i W_{ii}\,y_{ij}\big(\bar{z}_{ik} - y_{ij}\bar{f}_{jk}\big). \tag{15}$$

The first part of B^{surv} is the signal: the correlation, weighted by W, between gene j and the part of the working response that the other programs leave unexplained. The second part removes what the other genes of program k already explain.

7.3 Combined EBNM Problem

$$A_{f_{jk}} = A_{f_{jk}}^{\text{gen}} + w_S\,A_{f_{jk}}^{\text{surv}}, \qquad B_{f_{jk}} = B_{f_{jk}}^{\text{gen}} + w_S\,B_{f_{jk}}^{\text{surv}}. \tag{16}$$

EBNM Problem

$$(\bar{f}_{jk},\,\overline{f_{jk}^2}) \;\leftarrow\; \text{posterior under } g_{f_k} \text{ for } x_j = \frac{B_{f_{jk}}}{A_{f_{jk}}},\; s_j = \frac{1}{\sqrt{A_{f_{jk}}}}, \qquad j = 1,\ldots,p \text{ in sequence,}$$

then refit $\hat{g}_{f_k}$ by marginal likelihood from the sweep's (x_j, s_j).

Why the update is a sequential sweep over genes

$B_{f_{jk}}^{\text{surv}}$ depends on $\bar{f}_{j'k}$ for $j' \neq j$ through $\bar{z}_{ik}$. Exact coordinate ascent therefore updates one

gene at a time and refreshes $\bar{z}_{\cdot k}$ and $V^Z_{\cdot k}$ after each gene:

$$\bar{z}_{ik} \leftarrow \bar{z}_{ik} + y_{ij}\big(\bar{f}^{\text{new}}_{jk} - \bar{f}^{\text{old}}_{jk}\big), \qquad V^Z_{ik} \leftarrow V^Z_{ik} + y^2_{ij}\big(\text{Var}_q(f_{jk})^{\text{new}} - \text{Var}_q(f_{jk})^{\text{old}}\big).$$

The prior g_{f_k} is held fixed during the sweep, so each gene's posterior is a closed-form normal-means posterior, and it is refit once the sweep ends. At $w_S = 0$ the coupling disappears and the sweep equals the usual parallel EBMF update of f_k.

Precision-weighted combination. Because A and B are sums of a genomics and a survival part, the pseudo-observation is a precision-weighted average of a genomics-only estimate and a survival-only estimate:

$$x_j = \frac{A^{\text{gen}}\, x_j^{\text{gen}} + w_S A^{\text{surv}}\, x_j^{\text{surv}}}{A^{\text{gen}} + w_S A^{\text{surv}}}, \qquad x_j^{\text{gen}} = \frac{B^{\text{gen}}}{A^{\text{gen}}}, \quad x_j^{\text{surv}} = \frac{B^{\text{surv}}}{A^{\text{surv}}} \tag{17}$$

(all quantities for gene j and program k). The weight on the survival estimate is $\rho/(1+\rho)$ with $\rho = w_S A^{\text{surv}}/A^{\text{gen}}$. Section 13 works out the size of ρ.

Implementation Note

`fit_ssbmf_yfb()` calls `multimodal_yfb_update_F_mk()` (or its C++ version) with working weights $w_S W_{ii}$ and signal $w_S W_{ii} z_i^{-k}$, so the code computes exactly Eq. (16). A program with $\max_j A_{f_{jk}} < 10^{-8}$ receives no information from either likelihood. Its posterior would equal the prior and revive l_k and β_k from nothing, so the code sets $f_k = 0$ with a point-mass prior instead. This prunes the program while keeping K columns.

8 Update for τ: Noise Precision

τ appears only in the genomics term, since the Cox model uses the observed Y. Setting the derivative of Eq. (6) with respect to τ_j to zero gives the standard EBMF update

$$\hat{\tau}_j = \frac{n}{\sum_i \bar{R}^2_{ij}}, \tag{18}$$

with $\bar{R}^2_{ij}$ from Eq. (6) (posterior variances included).

9 Score Standardization

For any $c_k > 0$, the substitution

$$f_{\cdot k} \to f_{\cdot k}/c_k, \qquad l_{\cdot k} \to c_k\, l_{\cdot k}, \qquad \beta_k \to c_k\, \beta_k$$

leaves $LF^\top$ and $\boldsymbol{\eta} = YF\boldsymbol{\beta}$ unchanged. The reconstruction term cannot fix c_k, but the scale of $z_{\cdot k}$ changes the size of every survival quantity in the updates. Left free, the scale drifted from one iteration to the next. This feedback caused the April 2026 instability.

After initialization and at the end of every iteration, each program is rescaled so that its training scores have unit standard deviation:

$$c_k = \text{sd}_i\big(\bar{z}_{ik}\big), \qquad \bar{f}_{\cdot k} \leftarrow \bar{f}_{\cdot k}/c_k,\ \overline{f^2_{\cdot k}} \leftarrow \overline{f^2_{\cdot k}}/c_k^2,\ \bar{l}_{\cdot k} \leftarrow c_k \bar{l}_{\cdot k},\ \overline{l^2_{\cdot k}} \leftarrow c_k^2 \overline{l^2_{\cdot k}},\ \bar{\beta}_k \leftarrow c_k \bar{\beta}_k,\ \overline{\beta^2_k} \leftarrow c_k^2 \overline{\beta^2_k}. \tag{19}$$

The loading prior g_{f_k}, which is carried across sweeps, is transformed with f (a Laplace or exponential rate is multiplied by c_k, a normal variance divided by c_k^2). The priors on L and β are refit at every update and are scale families, so they need no transformation. A program whose scores have collapsed ($c_k < 10^{-6}$) is left unscaled.

What the scaling does and does not change

It selects one representative from a family of equivalent solutions. It changes neither the fitted reconstruction, the risk scores, nor the objective. It makes β_k a log hazard ratio per training standard deviation of program k's score, and it keeps the survival terms on a fixed scale between iterations. The code argument is `canonicalize = TRUE`. Earlier notes called the result the "canonical Z_F".

10 Initialization, Convergence, Pruning and Prediction

Initialization. $\bar{L}$ and $\bar{F}$ come from the rank-K SVD of Y, $\bar{L} = U_K D_K^{1/2}$ and $\bar{F} = V_K D_K^{1/2}$ (absolute values only when the prior is nonnegative), with $\overline{L^2} = \bar{L}^2$ and $\overline{F^2} = \bar{F}^2$. Then $\bar{\boldsymbol{\beta}} = \mathbf{0}$, $\hat{\tau}_j = 1/\widehat{\mathrm{var}}(y_{\cdot j})$, and the scaling of Section 9 is applied. An optional supervised start (`init_supervised = TRUE`) points program K along the null-model Cox score direction $Y^\top \boldsymbol{u}_0$.

Convergence. The monitored objective is Eq. (5) evaluated at the end of each iteration: the genomics term Eq. (6) at the updated τ, plus w_S times the survival term Eq. (9) re-expanded at the current $\bar{\boldsymbol{\eta}}$, minus the KL terms. The fit stops after iteration $b > 5$ when the relative change in this objective is below `tol`. The Taylor surrogate is re-expanded at every iteration and the priors are refit, so the objective is not guaranteed to increase at every step.

Pruning (optional, `prune = TRUE`). After convergence, for each program compute the change in the monitored objective when it is removed. If the largest change is nonnegative, remove that program and refit from the remaining programs' moments.

Prediction. For a new patient with expression $\boldsymbol{y}^*$, processed with the training transformations,

$$\hat{\eta}^* = \boldsymbol{y}^{*\top} \bar{F}\, \bar{\boldsymbol{\beta}} = \sum_k \bar{z}_k^*\, \bar{\beta}_k, \qquad \bar{z}_k^* = \boldsymbol{y}^{*\top} \bar{f}_{\cdot k}. \tag{20}$$

The training and test formulas are identical. No refit or posterior for the new patient's scores is needed (`predict_ssbmf_yfb()`).

11 Update Schedule: What Is Held Fixed, Rescaled, and Updated Together

Each update maximizes $\mathcal{F}$ over one block with everything else held at its current value. The table records the block, when it is computed, and what it treats as fixed. Any order of exact coordinate updates is valid coordinate ascent. The order affects the path and possibly the stationary point reached, not the update formulas.

Quantity	When computed	Held fixed / how updated
W_{ii}, z_i (Taylor)	Once, at the start of each outer iteration, at $\hat{\boldsymbol{\eta}} = Y\bar{F}\bar{\boldsymbol{\beta}}$	Frozen for the whole k-loop, even after $\bar{F}$ and $\bar{\boldsymbol{\beta}}$ change. Re-expanded at the end of the iteration for the objective only
$\bar{z}_{\cdot k'}$, $V_{\cdot k}^{Z}$	At the start of each program k	Reflect the programs already updated in this sweep (Gauss–Seidel over programs)
z_i^{-k}	At the start of each program k	Uses current $\bar{z}_{\cdot k'}$ and $\bar{\beta}_{k'}$ of the other programs
$q(\beta_k)$	One program at a time (scalar EBNM)	Uses the frozen W, z and current $\bar{Z}$. Prior refit at every update (per-program mode)
$q(l_k)$	All n patients together (one EBNM call)	Exact: given F and τ, the patients' scores are conditionally independent and share one precision. Prior refit at every update
$q(f_k)$	Genes one at a time, in order	Exact coordinate ascent: each gene sees the refreshed $\bar{z}_{\cdot k}$, $V_{\cdot k}^{Z}$. g_{f_k} fixed during the sweep, refit after it
Order within k	β_k, then l_k, then f_k	f_k uses the new $\bar{\beta}_k$, $\overline{\beta_k^2}$ and $\bar{l}_{\cdot k}$. (LB uses l_k, f_k, β_k)
τ	After the k-loop, all genes together	Closed form; genes are independent given L, F
Objective	After τ	Computed before rescaling. The rescaling leaves it unchanged
Score standardization	End of every iteration (and after initialization)	All programs at once; moments and g_{f_k} transformed (Section 9)
Stopping	After rescaling	Only after iteration 5

Two approximations in the schedule

1. The Taylor point is not refreshed inside the k-loop. Within an iteration the updates optimize a fixed quadratic surrogate, and the surrogate moves between iterations. This is the usual iteratively-reweighted scheme and the reason the objective can dip.

2. The prior on f_k is fixed during the gene sweep. Refitting it after each gene would make the sweep depend on gene order. Holding it fixed keeps each gene's update a closed-form posterior.

12 Complete Algorithm Summary

Supervised Matrix Factorization (YFB) — CAVI Algorithm

Inputs: $Y_{n\times p}$ (centered, standardized), $(t_i,\delta_i)_{i=1}^n$, number of programs K, survival weight w_S (default 1)
Initialize: $\bar{L},\bar{F}$ from the rank-K SVD of Y; $\bar{\boldsymbol{\beta}}=\mathbf{0}$; $\hat{\tau}_j=1/\widehat{\mathrm{var}}(y_{\cdot j})$; rescale so $\mathrm{sd}(Y\bar{f}_{\cdot k})=1$

For $b=1,2,\ldots$ **until convergence:**

1. Survival working quantities (once per outer iteration):

$$\hat{\boldsymbol{\eta}}=Y\bar{F}\bar{\boldsymbol{\beta}},\quad (u_i,W_{ii})=\mathrm{CoxTaylor}(\hat{\boldsymbol{\eta}},\boldsymbol{t},\boldsymbol{\delta}),\quad z_i=\hat{\eta}_i+u_i/W_{ii}$$

2. For $k=1,\ldots,K$:

$$\bar{z}_{ik'}=\textstyle\sum_j y_{ij}\bar{f}_{jk'},\quad V_{ik}^Z=\sum_j y_{ij}^2\,\mathrm{Var}_q(f_{jk}),\quad z_i^{-k}=z_i-\sum_{k'\neq k}\bar{z}_{ik'}\bar{\beta}_{k'}$$

(a) q_{β_k}, g_{β_k} (survival only):

$$A=w_S\textstyle\sum_i W_{ii}(\bar{z}_{ik}^2+V_{ik}^Z),\quad B=w_S\sum_i W_{ii}\,z_i^{-k}\,\bar{z}_{ik},\quad (\bar{\beta}_k,\overline{\beta_k^2})\leftarrow\mathrm{EBNM}(B/A,\,1/\sqrt{A})$$

(b) q_{l_k}, g_{l_k} (genomics only), with $R_{ij}^{-k}=y_{ij}-\sum_{k'\neq k}\bar{l}_{ik'}\bar{f}_{jk'}$:

$$A=\textstyle\sum_j\tau_j\overline{f_{jk}^2},\quad B_i=\sum_j\tau_j R_{ij}^{-k}\bar{f}_{jk},\quad (\bar{l}_{ik},\overline{l_{ik}^2})\leftarrow\mathrm{EBNM}(B_i/A,\,1/\sqrt{A})$$

(c) q_{f_k} (genomics + survival), genes $j=1,\ldots,p$ in sequence with g_{f_k} fixed:

$$A_j=\tau_j\textstyle\sum_i\overline{l_{ik}^2}\;+\;w_S\,\overline{\beta_k^2}\sum_i W_{ii}y_{ij}^2$$
$$B_j=\tau_j\textstyle\sum_i R_{ij}^{-k}\bar{l}_{ik}\;+\;w_S\Big[\bar{\beta}_k\sum_i W_{ii}y_{ij}z_i^{-k}-\overline{\beta_k^2}\sum_i W_{ii}y_{ij}(\bar{z}_{ik}-y_{ij}\bar{f}_{jk})\Big]$$
$$(\bar{f}_{jk},\overline{f_{jk}^2})\leftarrow\text{posterior under } g_{f_k}\text{ for }(B_j/A_j,\,1/\sqrt{A_j});\quad\text{update }\bar{z}_{\cdot k},\,V_{\cdot k}^Z$$

then refit g_{f_k} from $\{(B_j/A_j,\,1/\sqrt{A_j})\}_{j=1}^p$

3. Noise precision:

$$\bar{R}_{ij}^2=\Big(y_{ij}-\textstyle\sum_k\bar{l}_{ik}\bar{f}_{jk}\Big)^2+\sum_k\big(\overline{l_{ik}^2f_{jk}^2}-\bar{l}_{ik}^2\bar{f}_{jk}^2\big),\qquad\hat{\tau}_j=n/\sum_i\bar{R}_{ij}^2$$

4. Objective: $\mathcal{F}$ = genomics term + $w_S\times$ survival term (re-expanded at $Y\bar{F}\bar{\boldsymbol{\beta}}$, with $\mathrm{Var}_q(\eta_i)$) − KL terms
5. Score standardization: $c_k=\mathrm{sd}_i(Y\bar{f}_{\cdot k})$; $\bar{f}_{\cdot k}/c_k$, $\bar{l}_{\cdot k}\,c_k$, $\bar{\beta}_k\,c_k$ (second moments by $c_k^{\mp2}$); transform g_{f_k}
6. Convergence: stop when $b>5$ and $|\mathcal{F}^{(b)}-\mathcal{F}^{(b-1)}|/|\mathcal{F}^{(b-1)}|<\mathrm{tol}$

Prediction: $\hat{\eta}^*=\boldsymbol{y}^{*\top}\bar{F}\bar{\boldsymbol{\beta}}$

Where survival enters.

Update	Genomics	Survival	EBNM input
$q(\beta_k)$	–	$w_S\times$	scalar (x_k, s_k)
$q(l_k)$	✓	–	n-vector, common s
$q(f_k)$	✓	$w_S\times$	p-vector, sequential over genes
τ	✓	–	closed form

13 Balance of the Genomics and Survival Terms

Survival influences the factorization only through the F update, and there only through the share of the precision it contributes. By Eq. (17) that share is $\rho_{jk}/(1+\rho_{jk})$, with

$$\rho_{jk} = \frac{w_S A^{\text{surv}}_{f_{jk}}}{A^{\text{gen}}_{f_{jk}}} = \frac{w_S\,\overline{\beta_k^2}\sum_i W_{ii}\,y_{ij}^2}{\tau_j \sum_i \overline{l_{ik}^2}}. \tag{21}$$

Eq. (21) is exact and is computed in the code (`trace_supervision = TRUE`). The ratio is invariant to the scale substitution of Section 9, since numerator and denominator both scale by c_k^2. Its size can be read off approximately.

Numerator. With standardized expression and W_{ii} not strongly related to y_{ij}^2, $\sum_i W_{ii}y_{ij}^2 \approx \sum_i W_{ii} = D$ (Section 4). The numerator is about $w_S\,\overline{\beta_k^2}\,D$: it depends on the number of events and on β_k per standard deviation, and not on p.

Denominator. Let $d_k^2 = \|\bar{l}_{\cdot k}\|^2\|\bar{f}_{\cdot k}\|^2$ be the energy of program k's rank-one component and $v_k = d_k^2/(np)$ its share of the total variance of Y. When the programs are close to orthogonal and program k's scores are dominated by signal, $\bar{z}_{\cdot k} = Y\bar{f}_{\cdot k} \approx \bar{l}_{\cdot k}\|\bar{f}_{\cdot k}\|^2$. Unit standard deviation then means $\|\bar{l}_{\cdot k}\|\,\|\bar{f}_{\cdot k}\|^2 \approx \sqrt{n}$, and combining with the definition of d_k gives $\|\bar{l}_{\cdot k}\|^2 \approx d_k^4/n = n\,p^2 v_k^2$. Hence

$$\rho_{jk} \;\approx\; \frac{w_S\,\overline{\beta_k^2}\,(D/n)}{\tau_j\,p^2\,v_k^2}. \tag{22}$$

What Eq. (22) says

1. At fixed variance share, survival's precision share falls as $1/p^2$. Doubling the number of genes cuts it by four.
2. It grows as $1/v_k^2$. Survival has the most relative influence on low-variance programs, which are the programs supervision is meant to find, and almost none on dominant programs.
3. With the PDAC training data ($p \approx 2{,}200$, $D/n \approx 0.6$, $\overline{\beta_k^2} \approx 0.04$, $\tau_j \approx 1.5$, $v_k \approx 0.05$) and $w_S = 1$, Eq. (22) gives $\rho \approx 10^{-6}$. The trace on the real data measured 10^{-8} to 10^{-6} (DECISIONS.md, 2026-11-04). At that level the fitted loadings equal the unsupervised ones.
4. Matching the balance across p at fixed v_k needs $w_S \propto p^2$. Fixed weights of 40 or more broke convergence at $p = 2{,}000$ in simulation, through separation of the Cox model. The comparison document (`SSBMF-LB-vs-YFB-comparison/`) discusses ways to give survival more weight without that failure.

The approximation assumes signal-dominated scores and near-orthogonal programs. For a program near the detection threshold the noise part of $Y\bar{f}_{\cdot k}$ is not negligible and Eq. (21) should be used directly.

Comparison with the April 2026 claim. The April derivation concluded that the ratio is $O(1)$ because both precisions sum over n. It treated $\sum_i \overline{l_{ik}^2}$ as n times a fixed variance, but the size of l_k is tied to the size of f_k through the reconstruction and the score scale. Once that link is used, the ratio carries the factor $1/(p^2 v_k^2)$.

14 Strengthening Supervision Without Breaking the Fit

Section 13 shows that at $w_S = 1$ survival has almost no pull on the loadings of a dominant program at $p \approx 2{,}000$. This section lists the ways to change the balance and what each does to the derivation. None is adopted.

Why a large weight fails. The survival-only estimate in Eq. (17) solves, for program k, a weighted least squares problem $\beta_k Y \boldsymbol{f}_{\cdot k} \approx \boldsymbol{z}^{-k}$ with p unknowns and n observations. With $p > n$ it has exact solutions: some combination of genes orders the training survival times perfectly. As $\rho \to \infty$ the loadings follow that solution, the Cox fit separates, $|\bar{\beta}_k|$ grows without bound and the program detaches from expression structure. This matches the S-9 result that fixed weights of 40 or more lose convergence and the program. Any remedy has to keep the genomics term or the priors in control of the directions the Cox model alone cannot identify.

Approach	Change to the derivation	Status / risk
Weight w_S on the Cox term (current)	w_S in Eqs. (11) and (16); power posterior	S-9: $w_S \geq 40$ separates at $p = 2{,}000$
Survival-block weight (App. A)	w_S also multiplies $\mathrm{KL}(q_\beta \| g_\beta)$; β update at weight 1	Keeps the β prior in force; CV choice stable (S-13), criteria not all met
Choose w_S from data	Same updates; w_S by cross-validated partial log-likelihood, or a calibrated power (generalized Bayesian updating)	CV picks 1–10 in simulation; the candidate for the next real-data refit
Divide the genomics term by p	Equivalent to $w_S = p$ with the L, F KL terms unscaled	$\rho \propto 1/p$, still falls with p; already inside the separating range at $p \approx 2{,}000$
Normalize both losses and mix with α (DeSurv)	$(1-\alpha)\cdot\text{genomics}/c_G + \alpha\cdot\text{Cox}/c_S$ equals $w_S = \frac{\alpha}{1-\alpha}\frac{c_G}{c_S}$ up to an overall scale	Same family as w_S; DeSurv picks α by CV C-index
Fewer genes	$\rho \propto 1/p^2$, so a smaller gene set raises survival's share	A survival screen must sit inside CV to avoid reusing outcomes
Program-specific weights $w_{S,k}$	$w_{S,k}$ in program k's F and β updates	The Cox term couples programs, so this no longer maximizes one objective; heuristic only

What the balance analysis implies for the method

Because $\rho \propto 1/v_k^2$, a modest weight already acts on low-variance programs and leaves dominant programs to the genomics term. That is the regime where a joint fit can find what a two-step fit misses, and the near-threshold simulation (S-14) is where supervision showed a gain (validation C +0.156, 5 of 10 recoveries against 0). Choosing w_S by cross-validation targets that regime without moving into separation. A larger w_S cannot make survival dominate a high-variance program without first breaking the fit.

15 The Historical Configuration as a Special Case

The benchmark fits reported through October 2026 used `alpha_F = 0` with `canonicalize = FALSE`. The F update then mixed the two sources as $(1-\alpha_F)\cdot\text{genomics} + \alpha_F \cdot \text{survival}$, and at $\alpha_F = 0$ the survival part has weight zero.

At $\alpha_F = 0$ the fit is a two-step method

With no survival term in the F update, none of the L, F or τ updates uses (t, δ) or $\boldsymbol{\beta}$. The sequence of (q_L, q_F, τ) is the one an unsupervised EBMF fit would produce from the same start. The β update is an empirical Bayes Cox regression on the current scores. The result

is unsupervised EBMF followed by a shrinkage Cox fit, which is why the benchmark results did not differ from a two-step analysis. It remains a useful comparator.

How the historical weights differ from w_S.

	Historical (α, α_F)	Current (w_S)
F update	$(1-\alpha_F)A^{\text{gen}} + \alpha_F A^{\text{surv}}$; $\alpha_F > 0$ also down-weights genomics; B^{surv} incomplete	$A^{\text{gen}} + w_S A^{\text{surv}}$; genomics keeps weight 1; complete B^{surv}
β update	$A_{\beta_k} = \alpha \sum_i W_{ii} \bar{z}_{ik}^2 \cdot p$ (`per_p` divisor); no V^Z	$A_{\beta_k} = w_S \sum_i W_{ii} \overline{z_{ik}^2}$
Score in the Cox model	$Y\bar{f}_{\cdot k}/\|\bar{f}_{\cdot k}\|$, while the F update was derived for $Y\bar{f}_{\cdot k}$	$Y\bar{f}_{\cdot k}$ with sd = 1
Monitored objective	genomics and survival weighted by $(1-\alpha)$ and αp	the same w_S as the updates
Range	$\alpha, \alpha_F \in [0,1]$	$w_S \in [0,\infty)$

In both, the weight on β cancels in x_k and only changes s_k. The historical code also flips the sign of $\bar{\boldsymbol{\beta}}$ after fitting if the training C-index is below 0.5 (`sign_correction = TRUE`, the default). This is a post-processing heuristic, not part of the derivation, and the supervision studies set it to `FALSE`.

16 Extension to M Modalities

The single-modality model is the case $M = 1$ of a model with shared patient scores and modality-specific loadings (expression and methylation, for example):

$$Y_m = LF_m^\top + E_m, \quad E_{m,ij} \sim \mathcal{N}(0, \tau_{mj}^{-1}), \qquad \eta_i = \sum_k z_{ik}\beta_k, \quad Z = \sum_{m=1}^{M} Y_m F_m.$$

The full derivation is in `SSBMF-YFB-multimodal-model-derivations/`. Every update keeps the form derived here:

- $q(l_k)$: $A_{l_k} = \sum_m \sum_j \tau_{mj} \overline{f_{m,jk}^2}$ and $B_{l_{ik}} = \sum_m \sum_j \tau_{mj} R_{m,ij}^{-k} \bar{f}_{m,jk}$. Still genomics only.
- $q(f_{m,k})$: Eq. (16) with Y_m, τ_m and the joint score $\bar{z}_{\cdot k}$, which every modality's sweep refreshes. Each F_m is dual-source.
- $q(\beta_k)$: Eq. (11) with the joint score and $V_{ik}^Z = \sum_m \sum_j y_{m,ij}^2 \text{Var}_q(f_{m,jk})$.
- τ_m: Eq. (18) for each modality.
- Score standardization: $\text{sd}_i(\bar{z}_{ik}) = 1$ for the joint score.

With $M = 1$ every formula reduces to the one in this document.

New balance questions. The precision of l_k sums over all $p = \sum_m p_m$ features, so the modality with more features (methylation) dominates the shared scores. Survival's share of each loading precision falls as in Eq. (22), with the total p. Both point to modality-specific weights on the reconstruction terms, which need their own derivation and simulation check.

Convergence: what failed and what has changed since.

Earlier multimodal problem	Single-modality change that addresses it
Scale drift along $(a_k L_k, F_{mk}/a_k, a_k\beta_k)$; latent-moment stopping rules never fired (2026-09-18)	Score standardization each iteration (already in the multimodal fitter), and a scale-invariant objective
Stopping on relative change in reconstruction and predictor, with no recorded ELBO, so ELBO-based criteria could not be assessed (2026-10-15)	A monitored objective with the same weights as the updates and an ELBO stopping rule (Sections 3, 10)
Loading KL entered with the wrong sign through the shared helper (2026-10-15)	Fixed, with a test that recomputes the last objective from the returned moments
Real-data fits reached the sweep cap; refitted priors could have a support incompatible with the cached posterior KL (2026-10-15)	Prior held fixed during each sweep and refit once after it; a dead-program rule replaces revived zero programs
Free noise precisions ran away as reconstruction error grew (S-8)	Inputs standardized within platform; S-9 canonical fits converged in every case. A fixed or bounded τ remains a fallback
Incomplete B^{surv} in early single-modality code	The multimodal sweep already used the complete term; both fitters now share it

Recommended path.

1. Meet the single-modality acceptance criteria (recovery, false activity, convergence, external prediction) with this algorithm.
2. Build one fitter for M modalities whose $M = 1$ path reproduces `fit_ssbmf_yfb(canonicalize = TRUE)` to numerical tolerance. This is a direct test that the extension changes nothing for one block.
3. Carry over the monitored objective with w_S, the ELBO stopping rule, the dead-program rule and the KL tests. Record the full ELBO at every iteration.
4. Run two-modality simulations with planted programs in each modality and one shared prognostic program, then derive and test modality weights if methylation dominates L.

17 Differences from the April 2026 Derivations

ID	Location (`components/`)	Issue and correction	Here
Y1	`qF_supervised`, eq. BFsurv	B^{surv} removed the other programs but not the other genes of program k. Correct term subtracts $\overline{\beta_k^2}\sum_i W_{ii}y_{ij}(\bar{z}_{ik} - y_{ij}\bar{f}_{jk})$, and the update becomes a sequential sweep (found 2026-10-15).	Eq. (15)
Y2	`qF_supervised`, eqs. AF, BF	Convex mixing $(1-\alpha)/\alpha$ replaced by weight w_S on the Cox term; genomics weight 1.	Eq. (16)
Y3	`qF_supervised`, Sec. 4	"Scale imbalance resolved, ratio $O(1)$" is incorrect; the ratio scales as $1/(p^2 v_k^2)$.	Sec. 13
Y4	`qF_supervised`, Sec. 5	"EBNM shrinkage identifies the scale; no constraint needed." The free scale fed back into the survival terms; fixed by score standardization.	Sec. 9
Y5	`qBeta`, eqs. Abeta	Scores treated as observed: A_{β_k} used $\bar{z}_{ik}^2$. It should use $\overline{z_{ik}^2} = \bar{z}_{ik}^2 + V_{ik}^Z$.	Eq. (11)
Y6	`qBeta`, Sec. 2	Cox model used $Y\bar{F}D^{-1}$ (unit-norm columns), off the scale the F update was derived for.	Sec. 9
Y7	`ELBO`, term (II)	$\text{Var}_q(\eta_i)$ omitted the loading-uncertainty part $V_{ik}^Z\overline{\beta_k^2}$.	Eq. (9)
Y8	`ELBO` / code	Monitored objective weighted survival differently from the updates, and the loading KL entered with the wrong sign (fixed 2026-10-15).	Sec. 10

The $q(L)$ derivation (`qL_unsupervised`) and the τ update are unchanged.

A Options Tested and Not Adopted

Survival-block weight (`survival_block_weight = TRUE`). w_S multiplies $\text{KL}(q_\beta \| g_\beta)$ as well as the Cox term:

$$\mathcal{F}_1 = \mathbb{E}_q[\log p(Y \mid L, F, \tau)] - \text{KL}_L - \text{KL}_F + w_S\big[\mathbb{E}_q\,\ell_{\text{Cox}} - \text{KL}(q_\beta \| g_\beta)\big].$$

The L, F and τ updates are those of Eq. (5). For $w_S > 0$ the maximizer over (q_β, g_β) no longer depends on w_S, and the β update uses weight 1 in Eq. (11). This keeps the prior on β at full weight however large w_S is. In simulation (S-13), choosing w_S by cross-validated partial log-likelihood under this objective was stable but did not pass all registered criteria.

Joint Gaussian $q(\boldsymbol{\beta})$ (`beta_posterior = "joint"`). Replace $\prod_k q(\beta_k)$ by $\mathcal{N}_K(\boldsymbol{m}, \Sigma)$ with $G = w_S\big(\bar{Z}^\top W \bar{Z} + \text{diag}(\sum_i W_{ii}V_{ik}^Z)\big)$, $\boldsymbol{b} = w_S\bar{Z}^\top W\boldsymbol{z}$, $\Sigma = (G+\Lambda)^{-1}$, $\boldsymbol{m} = \Sigma\boldsymbol{b}$, and per-program prior variances by type-II maximum likelihood. For $K = 1$ this equals the normal-family EBNM update. In S-13 it gave the same results as the mean-field update.

Shared prior on $\boldsymbol{\beta}$ (`beta_prior_mode = "shared"`). One prior g_β for all programs, held fixed during the sweep and refit from the K pseudo-observations after it, in place of K priors each fit to one observation.